MODULE 4: MANUFACTURING PROCESSES, ENERGY CONVERSIONS AND MECHANICAL AUTOMATION

(CO: 4)

Complete Study Textbook, Step-by-Step Derivations & VIT FAT Question Bank

Aligned with Official VIT BACEE101 Course Page Syllabus, Question Bank, and Past FAT Papers

Scope of This Textbook (Official VIT Syllabus - All 13 Topics): (1) Metal Casting: Sand casting, patterns, moulding, defects. Metal Forming: Rolling, Forging, Extrusion, (2) Metal Joining: Arc welding, TIG, MIG, (3) Machining: Turning, Drilling, Milling, Grinding, Shaper, CNC, (4) Additive Manufacturing: FDM, (5) IC Engines: 2-stroke vs 4-stroke, CI vs SI, (6) Energy Sources: Thermal, Hydro, Nuclear power plants, (7) Renewable Energy: Solar, Wind, Fuel cells, (8) Refrigeration & AC, (9) Mechanisms: Kinematic links, pairs, Gruebler's equation, (10) Four-Bar Mechanism inversions, (11) Robotics: Structure & classifications, (12) Robotic Actuation: Hydraulic vs Pneumatic, and (13) Contemporary Topics.

1 VIT FAT Exam Blueprint & Master Principles

Module 4 is the largest mechanical/automation module, carrying **25 to 30 marks** in BACEE101 FAT:

1. **Manufacturing & Joining (10 Marks):** Sand casting process, patterns, gating system, casting defects (blowholes, shrinkage cavity), or TIG vs MIG arc welding comparison.
2. **Thermal Systems & Engines (10 Marks):** Working of 4-stroke SI (Petrol) vs CI (Diesel) engines, PV and TS indicator diagrams, or Thermal/Nuclear power plant layout.
3. **Mechanisms & Robotics (10 Marks):** Kinematic pairs, Gruebler's DOF mobility criterion ($F = 3(n-1) - 2j_1 - j_2$), Four-Bar inversions, and comparison between Hydraulic vs Pneumatic robotic actuators.

Topic	Governing Equation / Core Mechanism	VIT Exam Key Insight
Gruebler's Mobility Equation	$F = 3(n-1) - 2j_1 - j_2$	n = number of links, j_1 = lower pairs with 1 DOF, j_2 = higher pairs with 2 DOF.
Grashof's Law (Four-Bar)	$s + l \leq p + q$	At least one link can fully rotate 360° if sum of shortest and longest links $\leq$ sum of others.
SI vs CI Engines	Otto cycle (constant volume heat addition) vs Diesel cycle (constant pressure heat addition)	SI uses spark plug + carburetor/port fuel injection; CI uses fuel injector + high compression ($16:1 - 22:1$).
TIG vs MIG Welding	TIG: Non-consumable tungsten electrode + argon shield; MIG: Consumable wire spool electrode + CO_2/ argon	TIG produces cleanest aerospace welds; MIG offers high deposition rate for mass production.
COP of Refrigerator	$\text{COP}_R = \frac{Q_L}{W_{\text{in}}} = \frac{T_L}{T_H - T_L}$ (Carnot Limit)	Coefficient of Performance > 1; heat pump $\text{COP}_{\text{HP}} = \text{COP}_R + 1$.
Hydraulic vs Pneumatic	Hydraulic: Incompressible oil, high pressure (> 200 bar), high payload; Pneumatic: Compressible air ($6 - 10$ bar), fast, clean	Hydraulic delivers immense force but prone to fluid leaks; pneumatic is bouncy, fast, and spark-safe.

2 Manufacturing Processes: Casting, Forming, Joining & AM

2.1 Metal Casting & Sand Moulding

Metal casting pours molten metal into a refractory cavity to solidify into near-net-shape components:

- **Moulding Flask:** Consists of the upper **Cope** and lower **Drag**, separated by the **Parting Line**.
- **Pattern:** Replica of the final part, scaled up to incorporate pattern allowances:
  1. **Shrinkage Allowance:** Compensates for volumetric contraction during liquid-to-solid phase change and solid thermal cooling.
  2. **Draft Allowance:** Taper ($1^\circ - 3^\circ$) provided on vertical pattern faces to facilitate clean withdrawal from sand without destroying cavity walls.
  3. **Machining Allowance:** Extra stock for post-casting surface finishing.
- **Gating System:** Pouring basin → Down sprue → Runner → Ingate → Mould Cavity → Riser (reservoir of molten metal that feeds shrinking metal until solidification completes).
- **Casting Defects:** Blowholes (trapped gas/steam), Shrinkage cavities (inadequate riser feeding), Cold shut (incomplete fusion between two colliding metal streams), and Sand wash.

2.2 Metal Forming & Welding Processes

- **Metal Forming:** Bulk plastic deformation without material removal. **Rolling** passes metal between rotating rolls to reduce thickness; **Forging** hammers or presses billets into dies (hot or cold); **Extrusion** pushes metal through a shaped die orifice (like toothpaste).
- **TIG (Tungsten Inert Gas / GTAW):** Uses a non-consumable tungsten electrode under argon gas shielding. Filler rod is fed manually. Produces exceptionally clean, high-precision welds on stainless steel, aluminum, and titanium.
- **MIG (Metal Inert Gas / GMAW):** Uses a continuous consumable wire spool electrode automatically fed through the welding torch with shielding gas ($A\frac{r}{C}O_2$). Faster deposition, ideal for structural steel fabrication.

2.3 Additive Manufacturing: Fused Deposition Modelling (FDM)

A layer-by-layer 3D printing process where thermoplastic filament (PLA, ABS, PETG) is driven into an extrusion nozzle heated above glass transition temperature, extruded through a fine orifice (0.4 mm), and deposited onto a build plate along toolpaths generated by slicer software (G-code).

3 Energy Systems, Kinematics & Robotics

3.1 4-Stroke SI vs CI Internal Combustion Engines

1. **Suction Stroke:** SI admits air-fuel mixture through intake valve; CI admits pure air only.
2. **Compression Stroke:** SI compresses mixture to ratio $8:1 - 11:1$; CI compresses air to $16:1 - 22:1$, raising air temperature above the autoignition temperature of diesel fuel ($> 550^\circ$ C).
3. **Power (Expansion) Stroke:** SI ignites fuel via electric spark plug; CI injects atomized liquid diesel through high-pressure fuel injector, spontaneously igniting upon contact with hot compressed air.
4. **Exhaust Stroke:** Piston sweeps combustion exhaust gases out through open exhaust valve.

3.2 Kinematic Mechanisms & Gruebler's Mobility

A **Kinematic Link** is a rigid body. A **Kinematic Pair** connects two links permitting relative motion:

- **Lower Pair:** Surface contact (turning/revolute pair, sliding/prismatic pair, screw pair).
- **Higher Pair:** Point or line contact (cams, gears, ball bearings).

By Gruebler's Criterion, the Degree of Freedom (mobility F) of a planar mechanism is:

$$F = 3(n-1) - 2j_1 - j_2$$

For a standard Four-Bar Chain ($n = 4$ links, $j_1 = 4$ revolute hinges, $j_2 = 0$):

$$F = 3(4-1) - 2(4) - 0 = 9 - 8 = 1 \quad \text{(Single degree of freedom constraint)}$$

Four-Bar Mechanism Inversions:

1. **Crank-Rocker Mechanism:** Shortest link adjacent to fixed link rotates fully (360°), driving the opposite link to oscillate back and forth.
2. **Double-Crank (Drag-Link) Mechanism:** Shortest link is fixed; both connected links rotate complete 360° circles.
3. **Double-Rocker Mechanism:** Link opposite to shortest link is fixed; both connected links merely oscillate.